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Teorema de la función inversa

Teorema 1 (de la función inversa). *Sea $F\colon \Omega \subset \mathbb{R}^n \longrightarrow \mathbb{R}^n$ una aplicación $\mathcal{C}^1$ y $a \in \Omega$ tal que $DF(a)$ es invertible (i.e. $\det DF(a) \neq 0$), entonces*

(1) $\exists U, V \subset \mathbb{R}^n$ *abiertos con*

(a) $a \in U \subset \Omega \wedge F(a) \in V$. *(b)* F *inyectiva en* $U \wedge F(U) = V$.

(2) $\exists G\colon V \longrightarrow U$ *aplicación inversa de* $F\big|_U$ *(i.e.* $\forall x \in U : G(F(x)) = x$, *luego* G *es* $\mathcal{C}^1$*).*

Referenciado en

- `teo-fn-inversa-holomorfas`
- `teo-cartas-adaptadas-inmersion`
- `teo-difeomorfismo-local-iff-rango-es-dim`

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